

Rubrics

Q1. Gini Index Split Selection

[2.5 marks]

A parent node contains 10 samples:

6 positives, 4 negatives

Two candidate splits are given.

Step 1: Write the Gini formula

For a node with class proportions p_+ and p_- ,

$$G = 1 - p_+^2 - p_-^2$$

For a split,

$$G_{\text{split}} = \frac{n_L}{n}G(L) + \frac{n_R}{n}G(R)$$

Step 2: Compute parent Gini

$$\begin{aligned} G(P) &= 1 - \left(\frac{6}{10}\right)^2 - \left(\frac{4}{10}\right)^2 \\ &= 1 - \frac{36}{100} - \frac{16}{100} = 1 - \frac{52}{100} = 0.48 \end{aligned}$$

Step 3: Compute Split A

Left child: $4^+, 1^-$, total 5

$$\begin{aligned} G(L) &= 1 - \left(\frac{4}{5}\right)^2 - \left(\frac{1}{5}\right)^2 \\ &= 1 - \frac{16}{25} - \frac{1}{25} = 1 - \frac{17}{25} = \frac{8}{25} = 0.32 \end{aligned}$$

Right child: $2^+, 3^-$, total 5

$$G(R) = 1 - \left(\frac{2}{5}\right)^2 - \left(\frac{3}{5}\right)^2$$

$$= 1 - \frac{4}{25} - \frac{9}{25} = 1 - \frac{13}{25} = \frac{12}{25} = 0.48$$

Weighted Gini for Split A:

$$\begin{aligned} G_A &= \frac{5}{10}(0.32) + \frac{5}{10}(0.48) \\ &= 0.16 + 0.24 = 0.40 \end{aligned}$$

Step 4: Compute Split B

Left child: $5^+, 3^-$, total 8

$$\begin{aligned} G(L) &= 1 - \left(\frac{5}{8}\right)^2 - \left(\frac{3}{8}\right)^2 \\ &= 1 - \frac{25}{64} - \frac{9}{64} = 1 - \frac{34}{64} = \frac{30}{64} = 0.46875 \end{aligned}$$

Right child: $1^+, 1^-$, total 2

$$\begin{aligned} G(R) &= 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 \\ &= 1 - \frac{1}{4} - \frac{1}{4} = 0.5 \end{aligned}$$

Weighted Gini for Split B:

$$\begin{aligned} G_B &= \frac{8}{10}(0.46875) + \frac{2}{10}(0.5) \\ &= 0.375 + 0.1 = 0.475 \end{aligned}$$

Step 5: Compare the splits

Lower weighted Gini is better.

$$G_A = 0.40, \quad G_B = 0.475$$

Hence, Split A is better.

Marking scheme

- Correct Gini formula and parent computation: 0.5
- Split A child Ginis: 0.5
- Weighted Gini for Split A: 0.25
- Split B child Ginis and weighted Gini: $0.5 + .25$
- Correct conclusion: 0.5

Q2. OOB Error in Bagging

[2.5 marks]

True labels are:

$$x_1 : +, \quad x_2 : -, \quad x_3 : -, \quad x_4 : +$$

Step 1: Find OOB majority vote for each point

For x_1 : OOB predictions:

$$T_1 : +, \quad T_3 : +, \quad T_5 : -$$

Votes are $(+, +, -)$, so majority vote is $+$.

True label is $+$, so this is correct.

For x_2 : OOB predictions:

$$T_2 : -, \quad T_4 : -, \quad T_5 : +$$

Votes are $(-, -, +)$, so majority vote is $-$.

True label is $-$, so this is correct.

For x_3 : OOB predictions:

$$T_1 : -, \quad T_2 : +, \quad T_4 : -$$

Votes are $(-, +, -)$, so majority vote is $-$.

True label is $-$, so this is correct.

For x_4 : OOB predictions:

$$T_2 : -, \quad T_3 : +, \quad T_5 : -$$

Votes are $(-, +, -)$, so majority vote is $-$.

True label is $+$, so this is wrong.

Step 2: Compute OOB error

Out of 4 points, only x_4 is misclassified.

$$\text{OOB error} = \frac{1}{4} = 0.25$$

Hence,

$$\boxed{\text{OOB error} = 0.25}$$

Marking scheme

- Correct OOB error: 2.5
- In case step 2 is incorrect, but step 1 is correct, award 0.5 marks.

Q3. AdaBoost Weighted Error and α

[2.5 marks]

Given labels:

$$y = [+1, +1, -1, -1, +1]$$

Initial weights:

$$w_i^{(1)} = \frac{1}{5}, \quad i = 1, \dots, 5$$

Round 1 weak learner:

$$h_1(x) = [+1, -1, -1, +1, +1]$$

Round 2 weak learner:

$$h_2(x) = [+1, +1, -1, -1, -1]$$

Step 1: Compute weighted error in round 1

Compare $h_1(x)$ with true labels:

Example	y_i	$h_1(x_i)$	Correct/Wrong
x_1	+1	+1	Correct
x_2	+1	-1	Wrong
x_3	-1	-1	Correct
x_4	-1	+1	Wrong
x_5	+1	+1	Correct

Misclassified points are x_2 and x_4 .

Since each weight is $1/5$,

$$\epsilon_1 = \frac{1}{5} + \frac{1}{5} = \frac{2}{5} = 0.4$$

Hence,

$$\boxed{\epsilon_1 = 0.4}$$

Step 2: Compute α_1

AdaBoost coefficient:

$$\alpha_1 = \frac{1}{2} \ln \left(\frac{1 - \epsilon_1}{\epsilon_1} \right)$$

Substitute $\epsilon_1 = 0.4$:

$$\alpha_1 = \frac{1}{2} \ln \left(\frac{0.6}{0.4} \right) = \frac{1}{2} \ln(1.5)$$

Using $\ln(1.5) \approx 0.4055$,

$$\alpha_1 \approx \frac{1}{2}(0.4055) = 0.2027$$

So,

$$\boxed{\alpha_1 \approx 0.2027}$$

Step 3: Update weights for round 2

Correctly classified examples get factor $e^{-\alpha_1}$, and misclassified examples get factor e^{α_1} .

$$e^{-\alpha_1} \approx e^{-0.2027} \approx 0.8165$$

$$e^{\alpha_1} \approx e^{0.2027} \approx 1.2247$$

Unnormalized new weights:

$$w^{(2)} \propto \left[\frac{1}{5}(0.8165), \frac{1}{5}(1.2247), \frac{1}{5}(0.8165), \frac{1}{5}(1.2247), \frac{1}{5}(0.8165) \right]$$

$$= [0.1633, 0.2449, 0.1633, 0.2449, 0.1633]$$

Sum:

$$0.1633 + 0.2449 + 0.1633 + 0.2449 + 0.1633 = 0.9797$$

Normalized weights:

$$w^{(2)} \approx [0.1667, 0.25, 0.1667, 0.25, 0.1667]$$

Step 4: Compute weighted error in round 2

Round 2 predictions:

$$h_2(x) = [+1, +1, -1, -1, -1]$$

Compare with true labels:

Example	y_i	$h_2(x_i)$	Correct/Wrong	$w_i^{(2)}$
x_1	+1	+1	Correct	0.1667
x_2	+1	+1	Correct	0.25
x_3	-1	-1	Correct	0.1667
x_4	-1	-1	Correct	0.25
x_5	+1	-1	Wrong	0.1667

Only x_5 is misclassified, so

$$\epsilon_2 = 0.1667$$

Hence,

$\epsilon_2 = 0.1667$

Marking scheme

- Identify wrong examples and compute ϵ_1 : 1
- Compute α_1 : 0.5
- Correct weight update and normalization: 0.5
- Compute round 2 weighted error: 0.5

Q4. Weighted MSE in AdaBoost and Derivation of α [2.5 marks]

Suppose at round j , the predictor is updated as

$$F_j(x_i) = F_{j-1}(x_i) + \alpha h_j(x_i)$$

Define residual:

$$r_i = y_i - F_{j-1}(x_i)$$

Then the error after adding αh_j becomes

$$\begin{aligned} y_i - F_j(x_i) &= y_i - (F_{j-1}(x_i) + \alpha h_j(x_i)) \\ &= r_i - \alpha h_j(x_i) \end{aligned}$$

Step 1: Write weighted MSE loss

The weighted MSE loss is

$$L(\alpha) = \sum_{i=1}^N w_i (r_i - \alpha h_j(x_i))^2$$

Step 2: Expand the square

$$L(\alpha) = \sum_{i=1}^N w_i (r_i^2 - 2\alpha r_i h_j(x_i) + \alpha^2 h_j(x_i)^2)$$

Separate the terms:

$$L(\alpha) = \sum_{i=1}^N w_i r_i^2 - 2\alpha \sum_{i=1}^N w_i r_i h_j(x_i) + \alpha^2 \sum_{i=1}^N w_i h_j(x_i)^2$$

Step 3: Differentiate with respect to α

$$\frac{dL}{d\alpha} = -2 \sum_{i=1}^N w_i r_i h_j(x_i) + 2\alpha \sum_{i=1}^N w_i h_j(x_i)^2$$

Step 4: Set derivative to zero

$$-2 \sum_{i=1}^N w_i r_i h_j(x_i) + 2\alpha \sum_{i=1}^N w_i h_j(x_i)^2 = 0$$

Move terms:

$$2\alpha \sum_{i=1}^N w_i h_j(x_i)^2 = 2 \sum_{i=1}^N w_i r_i h_j(x_i)$$

Cancel 2:

$$\alpha \sum_{i=1}^N w_i h_j(x_i)^2 = \sum_{i=1}^N w_i r_i h_j(x_i)$$

Therefore,

$$\alpha^* = \frac{\sum_{i=1}^N w_i r_i h_j(x_i)}{\sum_{i=1}^N w_i h_j(x_i)^2}$$

Substitute $r_i = y_i - F_{j-1}(x_i)$:

$$\alpha^* = \frac{\sum_{i=1}^N w_i h_j(x_i)(y_i - F_{j-1}(x_i))}{\sum_{i=1}^N w_i h_j(x_i)^2}$$

Step 5: Special case when $h_j(x_i) \in \{-1, +1\}$

If $h_j(x_i) \in \{-1, +1\}$, then

$$h_j(x_i)^2 = 1$$

So denominator becomes

$$\sum_{i=1}^N w_i h_j(x_i)^2 = \sum_{i=1}^N w_i$$

If weights are normalized,

$$\sum_{i=1}^N w_i = 1$$

Hence,

$$\alpha^* = \sum_{i=1}^N w_i h_j(x_i)(y_i - F_{j-1}(x_i))$$

Marking scheme

- Correct weighted MSE formulation: 0.5
- Expansion of loss: 0.5
- Differentiate correctly: 0.5
- Solve correctly for α : 1